

CSCI 150 – Lab 2
Due: Wednesday, Week6
Submit to: Moodle

Objectives:

- Build circuits by using additional gates of **NAND, NOR, XOR, XNOR**, and **3-state Buffer**
- Verify the validity of Boolean algebra

Lab activities:

1. Login in to your Columbia College account
2. Right click on Start button, open Windows Explore
3. Create a new sub folder called **Lab2_yourFirstName** under your one drive
4. Start up LogicWorks 5
5. Read the follow the lab tutorial and build circuits based on the knowledge learnt in Lab1 and various Class Activities.
6. Save your practice to the file called **Lab2_a.cct, Lab2_b.cct, Lab2_c.cct**, under your Lab2 folder.
7. Observe the value changes at every I/O signal checkpoint (**Simulation /Show Value**).
8. Create Truth tables and verify the results with your circuits simulations.
9. For each circuit diagram, write your Name and ID, and the function Boolean equations beside the diagram.
10. Save screenshots and Truth Tables in **Lab2_report.docx** file.
11. Note: **Using LaTeX** for the Algebra expression whenever there is one to write.

1. Lab2 circuits

In LogicWorks, the 3-state buffer device provided, called **Buffer 1-T.S**, under the **Logic Simulation library**, is a kind of Active-“Low” tri-state buffer. See below illustrations:

http://www.electronics-tutorials.ws/logic/logic_9.html

Digital Buffer and the Tri-st...

File Edit View Favorites Tools Help

Active “HIGH” Tri-state Buffer

Symbol		Truth Table	
Enable	Data IN	A	Q
1	0	0	0
1	1	1	1
0	0	0	Hi-Z
0	1	1	Hi-Z

Read as Output = Input if Enable is equal to “1”

An **Active-high** Tri-state Buffer is activated when a logic level “1” is applied to its “enable” control line and the data passes through from its input to its output. When the enable control line is at logic level “0”, the buffer output is disabled and a high impedance condition, Hi-Z is present on the output.

Active “LOW” Tri-state Buffer

Symbol		Truth Table	
Enable	Data IN	A	Q
0	0	0	0
0	1	1	1
1	0	0	Hi-Z
1	1	1	Hi-Z

Read as Output = Input if Enable is NOT equal to “1”

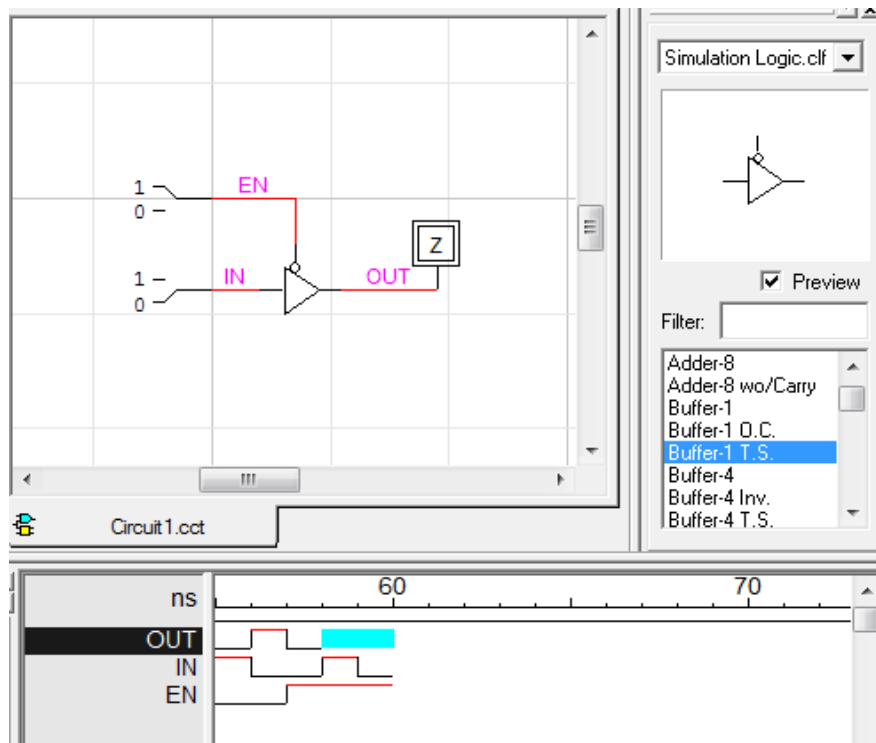
An **Active-low** Tri-state Buffer is the opposite to the above, and is activated when a logic level “0” is applied to its “enable” control line. The data passes through from its input to its output. When the enable control line is at logic level “1”, the buffer output is disabled and a high impedance condition, Hi-Z is present on the output.

Tri-state Buffer Control

The **Tri-state Buffer** is used in many electronic and microprocessor circuits as they allow multiple logic devices to be connected to the same wire or bus without damage or loss of data. For example, suppose we have a data line or data bus with some memory, peripherals, I/O or a CPU connected

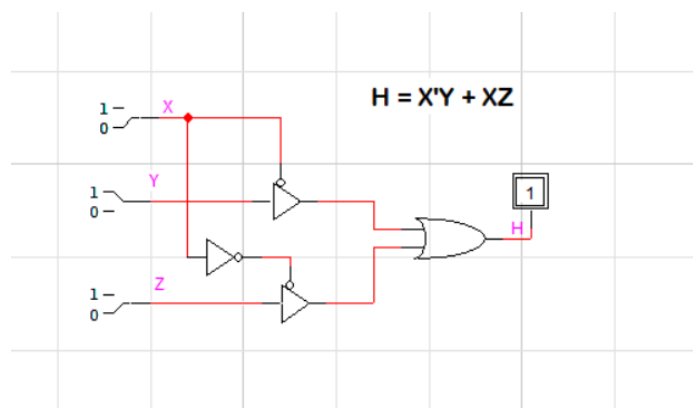
- Use a Buffer-1.T.S to simulate $Out = \overline{EN} \cdot IN$
 - ✓ Select “**Buffer-1 T.S**” from the **Logic Simulation** library
 - ✓ Follow the illustration below.
 - ✓ Save it as **Lab3_a.cct**.

Note: the enabler EN is inverted (with a bubble), which means when $EN = 0$, the buffer will be enabled. See the following screenshot.



b. According to your Lab3_a.cct, implement the function $H = \overline{X} \cdot Y + X \cdot Z$, using **two three-state buffers** and **one inverter**.

- 1) Draw a **Buffer-1 T.S** for $H1 = \overline{X} \cdot Y$
- 2) Draw a second **Buffer-1 T.S**, with one **inverter input** for $H2 = X \cdot Z$
- 3) Using a OR gate to interconnect 1) and 2).
- 4) Save the circuit as **lab2_b.cct**



- c. Construct an **exclusive-OR gate**, such as $F = X \oplus Y$, by interconnecting **two (Low) three-state buffers** (i.e. **Buffer-1 T.S**). Show the truth table to proof that both are equal. Save the circuit as **lab2_c.cct**
- d. Make sure you always **write your name on any of your circuit screens**. Save the **circuits (a, b, c)** and simulations with the resulting **truth tables (a, b, c)** in your lab report. **lab2_report.docx**

2. Submit your work:

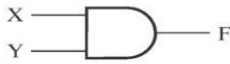
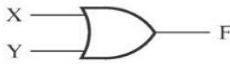
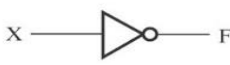
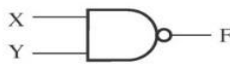
- Make sure you have created the following compressed file including 4 individual files:
 - Lab2_yourID_yourName.zip (compressed file)
 - ✓ Lab2_a.cct
 - ✓ Lab2_b.cct
 - ✓ Lab2_c.cct
 - ✓ Lab2_report.docx (include circuit diagrams and truth tables)
- Submit the zip file to Moodle.

3. References: Logical Symbols and Truth tables

1) Truth Table samples:

index	X Y Z	H1 = ...	H2=...	H= ...	$F = X \oplus Y = \dots$
0	0 0 0				
1	0 0 1				
2	0 1 0				
3	0 1 1				
4	...				
5	...				
6	...				
7	...				

2) List of the logical symbols and their truth tables:

Name	Distinctive-Shape Graphics Symbol	Algebraic Equation	Truth Table															
AND		$F = XY$	<table><tr><th>X</th><th>Y</th><th>F</th></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td></tr></table>	X	Y	F	0	0	0	0	1	0	1	0	0	1	1	1
X	Y	F																
0	0	0																
0	1	0																
1	0	0																
1	1	1																
OR		$F = X + Y$	<table><tr><th>X</th><th>Y</th><th>F</th></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td></tr></table>	X	Y	F	0	0	0	0	1	1	1	0	1	1	1	1
X	Y	F																
0	0	0																
0	1	1																
1	0	1																
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NOT (inverter)		$F = \overline{X}$	<table><tr><th>X</th><th>F</th></tr><tr><td>0</td><td>1</td></tr><tr><td>1</td><td>0</td></tr></table>	X	F	0	1	1	0									
X	F																	
0	1																	
1	0																	
NAND		$F = \overline{X \cdot Y}$	<table><tr><th>X</th><th>Y</th><th>F</th></tr><tr><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td></tr></table>	X	Y	F	0	0	1	0	1	1	1	0	1	1	1	0
X	Y	F																
0	0	1																
0	1	1																
1	0	1																
1	1	0																
NOR		$F = \overline{X + Y}$	<table><tr><th>X</th><th>Y</th><th>F</th></tr><tr><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>0</td></tr></table>	X	Y	F	0	0	1	0	1	0	1	0	0	1	1	0
X	Y	F																
0	0	1																
0	1	0																
1	0	0																
1	1	0																
Exclusive-OR (XOR)		$F = X\overline{Y} + \overline{X}Y$ $= X \oplus Y$	<table><tr><th>X</th><th>Y</th><th>F</th></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td></tr></table>	X	Y	F	0	0	0	0	1	1	1	0	1	1	1	0
X	Y	F																
0	0	0																
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Exclusive-NOR (XNOR)		$F = \overline{XY + \overline{X}\overline{Y}}$ $= X \oplus Y$	<table><tr><th>X</th><th>Y</th><th>F</th></tr><tr><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td></tr></table>	X	Y	F	0	0	1	0	1	0	1	0	0	1	1	1
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